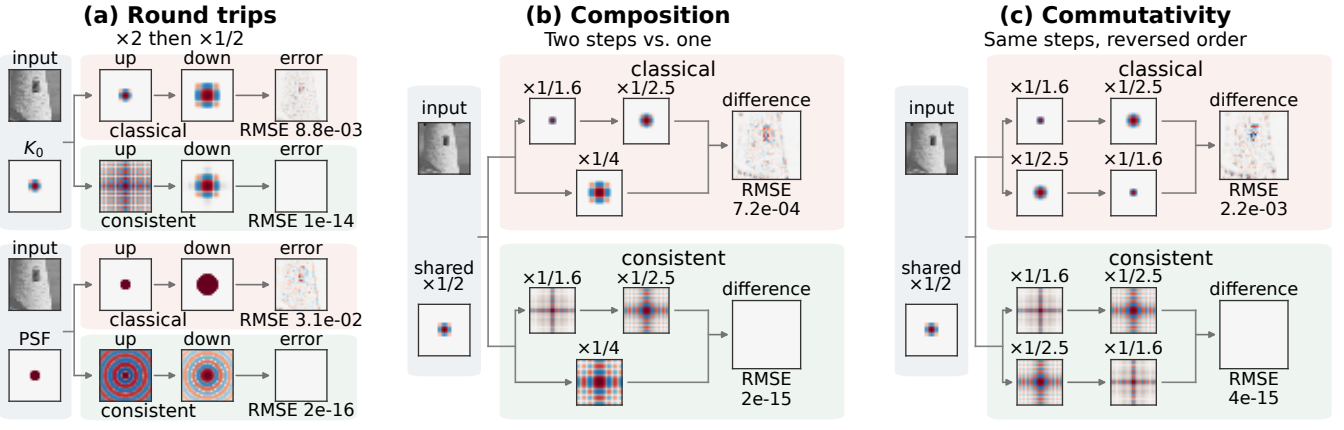

(a) Round trips
×2 then ×1/2
input
K_0
up
down
error
classical
RMSE 8.8e-03
consistent
RMSE 1e-14
PSF
RMSE 3.1e-02
RMSE 2e-16
(b) Composition
Two steps vs. one
shared
×1/2
×1/1.6
×1/2.5
×1/4
difference
RMSE
7.2e-04
2e-15
(c) Commutativity
Same steps, reversed order
4e-15
2.2e-03